Fiscal Year:	FY 2005	**Task Last Updated:**	FY 08/18/2005
PI Name:	Marston, Philip L. Ph.D.		
Project Title:	Passive and Active Stabilization of Liquid Bridges in Low Gravity		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Dynamics and stability		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Pullman	**State:**	WA
Zip Code:	99164-2814	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	03/10/2000	**End Date:**	03/09/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Thiessen, David B. (Washington State University) Thiessen, David B. (Washington State University)		
Grant/Contract No.:	NAG3-2378		
Performance Goal No.:			
Performance Goal Text:			
	There are two general types of stabilization methods being investigated. These use feedback control of radial electrostatic stresses or the acoustic radiation pressure of high frequency sound waves. In the electrostatic approach the spatial distribution of the Maxwell stress on the bridge is actively adjusted so as to counteract the growth of the most unstable mode. This is achieved by optically sensing the shape of the bridge and electronic control of the voltage applied to an array of ring-shaped electrodes that are centered on the bridge. Acoustic stabilization of liquid bridges is investigated with appropriately designed ultrasonic fields. One approach involves the selection of the acoustic field such that a local variation in the bridge radius automatically produces a compensating change in the local acoustic radiation pressure. In another approach the spatial distribution of the acoustic radiation pressure on the bridge is actively adjusted so as to counteract the growth of the most unstable mode. This is achieved by optically sensing the shape of the bridge as in the electrostatic case. Active reduction of the response of liquid bridges to g-jitter should also be possible with an		

Task Description:	appropriate feedback algorithm. The bridges for investigation are primarily liquid bridges in air, however some bridges surrounded by an immiscible host liquid are also studied. The primary objective is to stabilize liquid bridges against break up from the Rayleigh-Plateau instability. A secondary objective is to reduce the response of liquid bridges to g-jitter using active damping and modification of the natural frequency of bridge oscillations. The research should give improved capabilities to manage fluids in low-gravity by achieving stabilization beyond the Rayleigh-Plateau limit and by reducing the response of liquid bridges to g-jitter. This research may have application to float-zone crystal growth processes or to the containerless management of liquids and the production and control of liquid drops. Long liquid bridges should be beneficial to float zone processing because it is possible to reduce thermal gradients in the liquid as well as in the solid. Reducing thermal stresses in newly formed crystals can be important to the crystal quality. The ability to provide active damping may have significant benefits for environments having significant g-jitter.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The stabilization and active damping methods developed ate not limited in applicability to liquid bridges in reduced gravity since they may be scaled to earth-based situations involving free liquid interfaces.
Task Progress:	Recent progress is covered in the Final Report for this grant submitted in June 2005 and in the thesis: Wei Wei "Capillary Bridge Stability and Dynamics: Active Electrostatic Stress Control and Acoustic Radiation Pressure," Ph. D thesis Washington State University (May, 2005). Recent progress is also described in the following peer-reviewed publications: D. B. Thiessen, W. Wei, and P. L. Marston, "Active electrostatic control of liquid bridge dynamics and stability," Annals of the New York Academy of Sciences 1027, 495-510 (2004). W. Wei, D. B. Thiessen, and P. L. Marston, "Enhanced damping of capillary bridge oscillations using velocity feedback," Physics of Fluids 17, 032105 (2005) (11 pages). Wei Wei, David B. Thiessen, and Philip L. Marston "Control of the natural frequency of the (2,0) mode of liquid bridges using active electrostatic fields" Physics of Fluids 17, 071702 (2005) (4 pages).
Bibliography Type:	Description: (Last Updated: 06/21/2021)
Abstracts for Journals and Proceedings	Wei, W.; Thiessen, D. B.; and Marston, P. L. "Control of the frequency of the (2,0) mode of liquid bridges using active electrostatic fields" APS 57th Division of Fluid Dynamics Mtg. Bulletin of the American Physical Society 49 (9) 140-141 (2004) , Nov-2004
Articles in Peer-reviewed Journals	Thiessen, D. B.; Wei, W.; and Marston, P. L. "Active electrostatic control of liquid bridge dynamics and stability" Annals of the New York Academy of Sciences 1027, 495-510 (2004) , Nov-2004
Articles in Peer-reviewed Journals	Wei, W.; Thiessen, D. B.; and Marston, P. L. "Enhanced damping of capillary bridge oscillations using velocity feedback" Physics of Fluids 17, 032105 (2005) (11 pages) , Mar-2005
Articles in Peer-reviewed Journals	Wei, W.; Thiessen, D. B.; and Marston, P. L. "Control of the natural frequency of the (2,0) mode of liquid bridges using active electrostatic fields" Physics of Fluids 17, 071702 (2005) (4 pages). , Jul-2005
Dissertations and Theses	Wei Wei "Capillary Bridge Stability and Dynamics: Active Electrostatic Stress Control and Acoustic Radiation Pressure" Ph. D thesis Washington State University (2005) May-2005